

Neural network-based reconstruction of steady-state temperature systems with unknown material composition

一、論文的挑戰

在材料分布、熱源位置、邊界條件都未知的情況下，只靠少量溫度量測點，重建完整 2D 溫度場。傳統方法的困難：

1. 實際系統不可能到處放感測器。
2. 有些內部材料熱傳導率不同，但使用者不一定知道材料分布。
3. 熱源位置與大小也可能未知。
4. 若用 FEM 等物理模擬，需要知道材料、熱源、邊界條件，否則很難直接算。
5. 稀疏感測時，尤其只在邊界放感測器，內部 hotspot 很難被發現。

二、論文的假設

1. 系統可近似為 2D 穩態熱傳導問題。
2. 輸入資料是部分溫度點，也就是被 mask 的溫度場。
3. 不提供材料分布、熱源位置、熱源大小、邊界條件給模型。
4. 訓練資料由 FEM 模擬產生，作為 ground truth。
5. 實際應用中，雖然不知道完整材料分布，但系統的材料種類與幾何範圍仍有大致限制。

三、提出解決的方案

Spatial Propagator + Physics-informed Fully Convolutional Autoencoder

1. 輸入部分溫度場:
 - (1) Scattered sampling: 隨機在平面上取一些溫度點
 - (2) Ring sampling: 只取邊界一圈溫度
 - (3) Grid-edge sampling: 模擬邊界上離散感測器
→ **Grid-edge** 最接近真實情況
2. Spatial Propagator: 這個模組先把已知溫度資訊往空間內部傳播。先讓邊界或少量量測點的資訊擴散到整個影像，再交給後面的 U-Net 類模型重建。
3. U-Net-like Autoencoder: 從不完整溫度圖中學習完整溫度場的空間分

布。

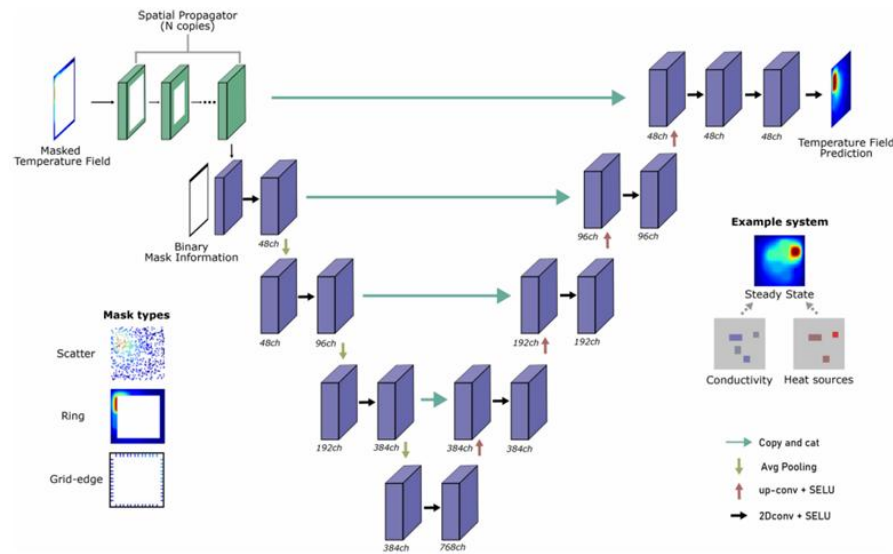


Fig. 2. Illustration of the U-Net-based neural network architecture designed for reconstructing 2D temperature fields from partially observed data. The process initiates with a 'Spatial Propagator' that pre-processes the input masked temperature fields, followed by a U-Net for the prediction of the complete temperature distribution. The figure also presents three distinct masking approaches used in our experiments: 'Random', 'Ring', and 'Grid-edge', alongside an exemplar system showcasing a randomly distributed conductivity and heat-source field, and its resulting steady-state temperature field as simulated for dataset creation.

4. 加入 Physics-informed loss: Loss 不只看溫度誤差，也看物理合理性。

$$L_{total} = \alpha_T \cdot L_T + \alpha_G \cdot L_G + \alpha_H \cdot L_H,$$

- $L_T = \frac{1}{N} \sum_{pixels} \|T - T'\|^2$ measures the discrepancy in temperature fields.
- $L_G = \frac{1}{N} \sum_{pixels} \|\nabla T - \nabla T'\|^2$ focuses on the gradient of temperature fields.

四、評估環境：模擬+實體實驗

1. 模擬：

(1) 使用 FEM 產生 2D 穩態溫度場資料：

- 系統大小: 50mm×50mm
- 資料解析度: 50×50=2500 pixels
- 材料: Copper、Epoxy、Silicon、FR4、AlO、Aluminum，
- 資料量: 775 system→資料旋轉增強 3100(775 × 4)
- Train/Test: 80%/20%

2. 實體實驗: PCB 測試

驗證模型雖然只用 2D 模擬資料訓練，但能不能應用到真實 PCB 溫度重建。結果顯示可以合理重建，但因為真實 PCB 是 3D 系統，所以仍有誤差。

(1) 實驗材料：

- SMD prototyping board

- b. FR4 PCB
- c. Copper pads
- d. SMD resistors 作為熱源
- e. 熱像儀 InfraTec VarioCam HD 量測溫度場

五、主要結果

1. 即使 Grid-edge sampling 只有3.12%的取樣比例，模型仍可達到約1.1%的平均相對誤差。→用少量邊界感測器，也可以重建完整溫度場。

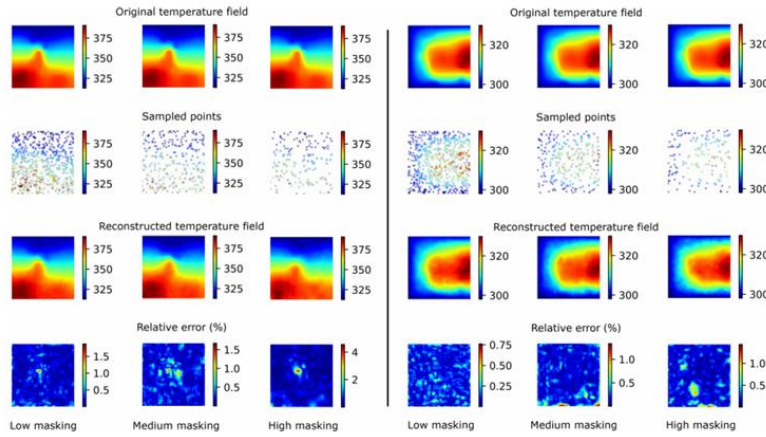


Fig. 3. Two examples of the reconstructed temperature fields for the scattered sampling method, for three different masking ratios. The rows are, from top to bottom, the ground truth temperature, the sampled temperature values, the reconstructed temperature field and the relative error in percent (see Eq. 2). Each of the three columns per system corresponds to different masking ratios, which amounts to a different sampling ratio as described in the text.

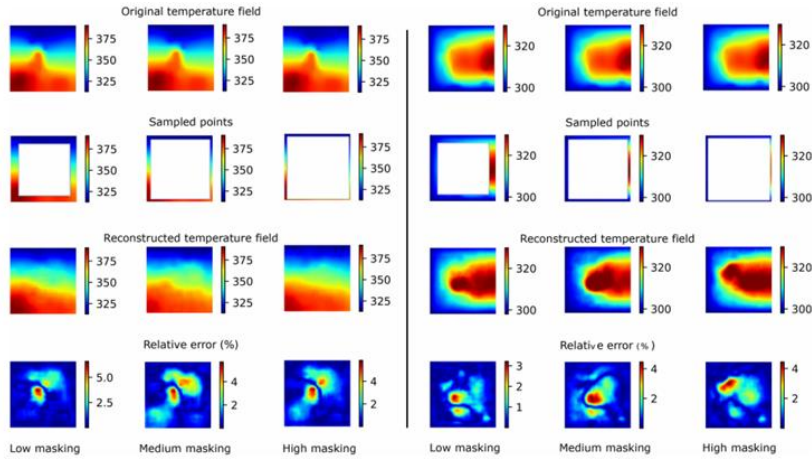


Fig. 4. Two examples of the reconstructed temperature fields for the ring sampling method, for three different masking ratios. The rows are, from top to bottom, the ground truth temperature, the sampled temperature values, the reconstructed temperature field and the relative error in percent (see Eq. 2). Each of the three columns per system corresponds to different masking ratios, which amounts to a different sampling ratio as described in the text.

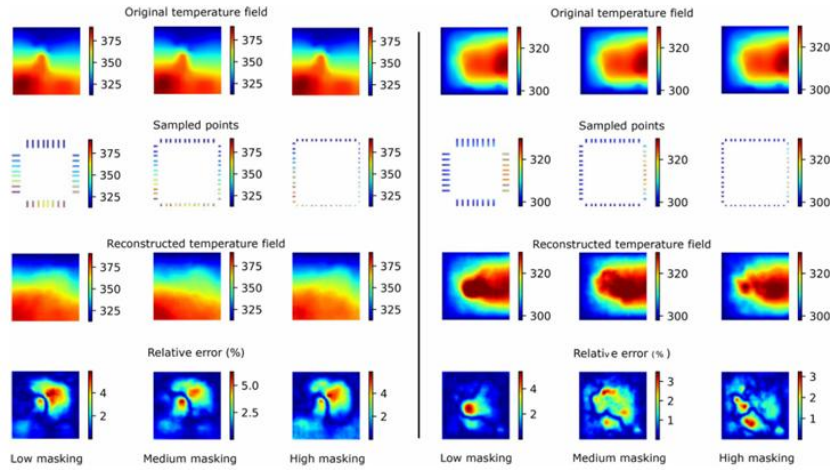


Fig. 5. Two examples of the reconstructed temperature fields for the Grid-edge sampling method, for three different masking ratios. The rows are, from top to bottom, the ground truth temperature, the sampled temperature values, the reconstructed temperature field and the relative error in percent (see Eq. 2). Each of the three columns per system corresponds to different masking ratios, which amounts to a different sampling ratio as described in the text.

	Scattered low masking (%)	Scattered medium masking (%)	Scattered high masking (%)
No noise	0.20	0.20	0.24
With noise	0.16	0.20	0.24
	Ring low masking (%)	Ring medium masking (%)	Ring high masking (%)
No noise	0.60	0.92	1.00
With noise	0.60	1.14	0.91
	Grid-edge low masking (%)	Grid-edge medium masking (%)	Grid-edge high masking (%)
No noise	0.78	0.97	1.11
With noise	0.82	0.99	1.09

Table 2. Average relative errors for different sampling patterns and masking levels.

2. 與傳統空間內插方法 Kriging 比較:

- (1) Kriging: 結果較平滑，但容易低估最高溫。
- (2) Neural Network: 較能抓到最高溫與熱點趨勢，但需要訓練資料

最高溫比平均溫度更重要，因為 hotspot 可能造成元件損壞。所以 NN 方法比 Kriging 更適合 thermal monitoring。

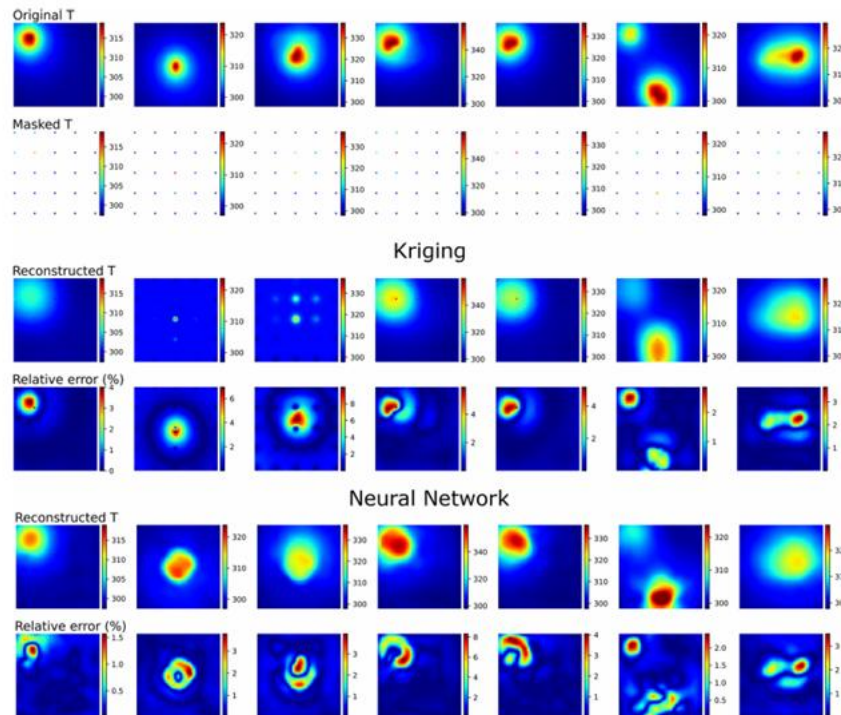


Fig. 9. Comparison between Kriging and our neural network approach for several examples of reconstructed temperature fields from measurement data. A regular scattered sampling pattern is used. The rows are, from top to bottom, the ground truth temperature, the sampled temperature values, the reconstructed temperature field and the relative error in percent for the two different approaches respectively (see Eq. 2).

六、論文限制

1. 目前主要是 2D 模型，真實系統通常是 3D。
2. 訓練資料是 FEM 模擬產生，實際系統若差異太大，效果可能下降。
3. 對非常局部的 hotspot 偵測能力不足。
4. 如果只在邊界取樣，且熱沒有明顯傳到邊界，模型會失敗。
5. 模型不是 general-purpose，不能保證適用所有熱系統。
6. 需要先設計足夠代表性的訓練資料集。
7. 實驗 PCB 中的對流、輻射、厚度方向散熱沒有完整納入 2D 模型。

七、總結問題

本文解決的是在材料、熱源與邊界條件未知的情況下，如何利用少量溫度感測點重建完整穩態溫度場的問題。

八、跟你的研究方向的關聯